

Joe (Hoang) Do

703-397-6188 | joedo2910@vt.edu | linkedin.com/in/hoanglehuydo/ | github.com/notjoedo | joedo.dev

EDUCATION

Virginia Tech

B.S. in Computational Modeling and Data Analytics, Minor in Computer Science, Math

GPA: 3.66/4

Blacksburg, VA

Aug. 2023 – May 2027

TECHNICAL SKILLS

Software Skills: Python, Java, JavaScript/TypeScript, C, C#, HTML/CSS, R, SQL, React, Next.js, FastAPI, Django, Flask, Robotic Process Automation (openRPA, UIPath), Parallel Computing (CUDA, OpenMP), Splunk (SPL), Grafana, Digital Twins

Developer Tools: Git, Docker, AWS, Azure, PostgreSQL, Vercel, Figma

EXPERIENCE

AI & IoT Solutions Intern

SAS

May 2026 – Present

Cary, NC

- Created real-time streaming pipeline from SAS Event Stream processing (ESP) to RAM (Retrieval Agent Manager), **enabling low-latency delivery of contextualized event data** for downstream intelligent processing.
- Engineered real-time anomaly detection in SAS ESP using pattern windows, stateful rule-based logic, and an SVDD (Support Vector Data Description) model to identify faulty conditions, enabling **immediate alerting, automated classification, and AI-driven downstream analysis** via a RAM-based retrieval-augmented pipeline.
- Developed a UE5-based digital twin system to simulate and visualize real-time system behavior, enabling **interactive control, improved system observability, and more efficient parameter tuning** through a synchronized front-end interface.

AI Integration & Full-Stack Software Engineering Intern

Centennial Technologies

May 2025 – July 2025

Leesburg, VA

- Architected and deployed a **Retrieval-Augmented Generation (RAG)** system for Agentic-AI platform, utilizing Azure services (**Blob Storage, AI Foundry, Functions, VM**) to automate data sourcing, enabling users to query FDA approval document data in under 3 seconds.
- Engineered a hybrid Agentic-AI solution by developing a **Robotic Process Automation (RPA)** bot using **Optical Character Recognition (OCR)** and **Azure services** to automate data entry/extraction for XFA files, eliminating manual bottlenecks and enabling Agentic-AIs to autonomously execute end-to-end workflows by 90%.
- Designed UI/UX from wireframes and developed a modern frontend (**React, TypeScript, Tailwind**) that provided an intuitive and accessible interface, passing WCAG tests for Agentic-AI document processing system.

Lead SWE Undergraduate Researcher

Virginia Tech GrayUR's Phased Array Microphonics Team

Aug 2024 – December 2025

Blacksburg, VA

- Directed a 5-member cross-disciplinary team, serving as the technical leader for the integration of custom algorithms with phased array hardware and utilizing **Trello** to manage dependencies and track complex project deliverables.
- Enhanced project's analytical capabilities by researching, developing, and integrating a **custom PyTorch model**, improving data prediction accuracy by 55% and automated a previously manual analysis process.
- Established model serving and data management protocols (**SQL/NoSQL**) to ensure the scalability and reliability of the PyTorch prediction service, drastically reducing latency and facilitating real-time data access.

Splunk Dashboard Data Analytics Developer Intern

Virginia Tech Division of Information Technology

Aug 2024 – May 2025

Blacksburg, VA

- Enhance system monitoring by translating stakeholder needs into **15+ analytical dashboards using Splunk SPL**, which reduced the identification time for critical security events by 90%.
- Detect system vulnerabilities in real-time by developing a smart alert system in Splunk that successfully **identified over 100 critical errors and security breaches daily**, leading to 50% decreased incident response time.
- Optimized data ingestion performance by architecting new log forwarding configurations and data validations across 10+ data sources, **resulting in a 30% reduction in search latency and improved system reliability**.